

Statistics for Data Science

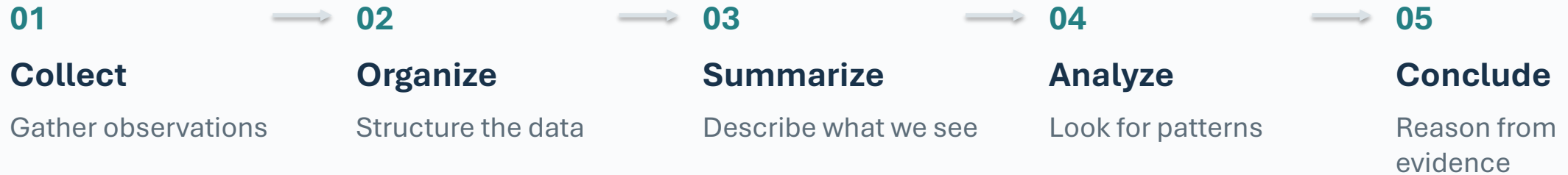
From data to information and decisions

Today: populations and samples, variables, center, and spread



What Is Statistics?

Statistics is the science of learning from data.

**Key idea**

Data alone do not automatically provide useful information.

Example scores: 72, 81, 95, 68, 87, 91, 84, 77, 88, 52

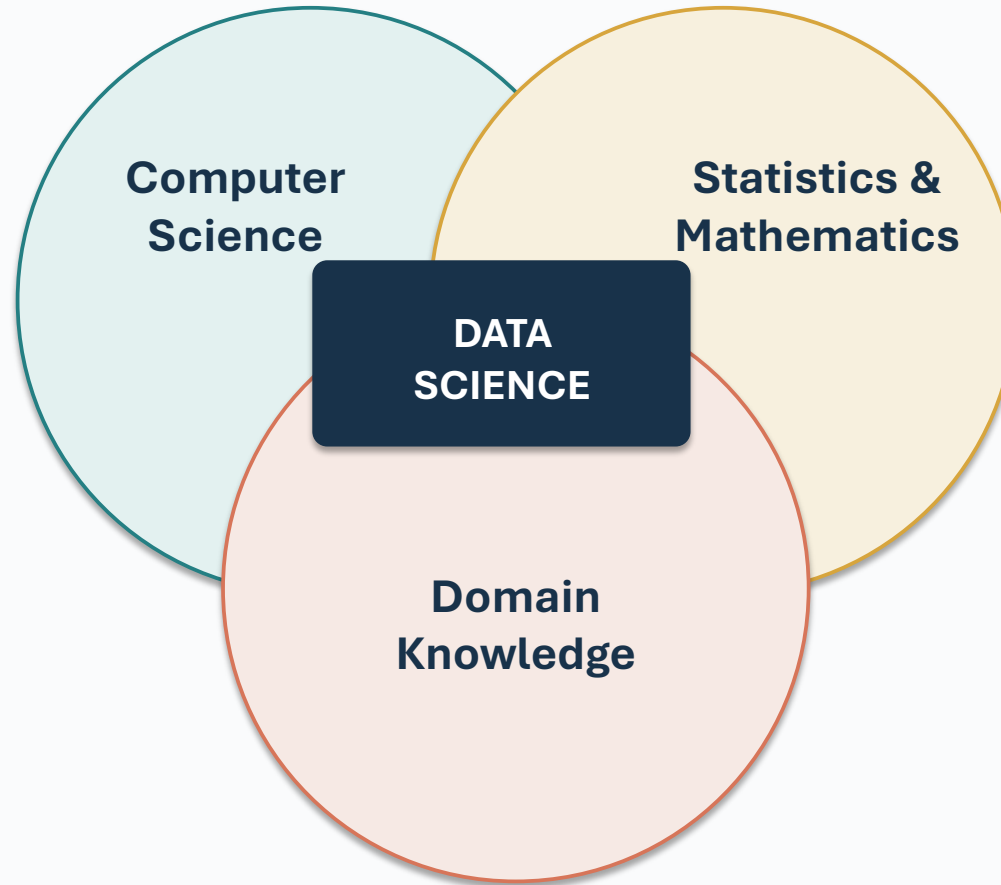
What was typical? How much did scores vary? Is 52 unusual?



**Decision-
Making**

Statistics Is One Part of Data Science

Data science combines three areas of expertise.



Computer science

Programming
Algorithms
Databases

Statistics

Understand variability
Identify patterns
Quantify uncertainty

Domain knowledge

Ask useful questions
Interpret results

Is Recommendation System B Actually Better?

An online store compares average spending for users who see two systems.

\$52.30

System A

vs.

\$54.10

System B

Observed gap

+\$1.80

Possible explanations

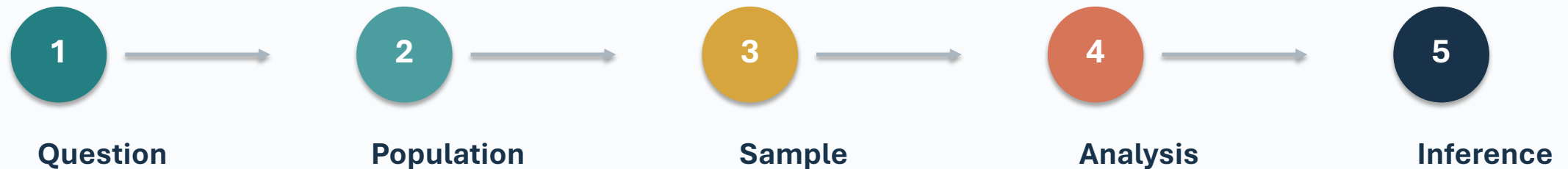
- **A is better** The recommendation changed spending.
- **Random variation** The gap may come from chance.
- **Different users** The groups may differ before exposure.
- **Data quality** Sample sizes or outliers may distort the averages.

Statistics helps separate signal from noise.

A Typical Statistical Investigation

Question: Do UNCG students spend more than 3 hours per day on social media?

- Which observation is the most occurring?
- Is there a difference between the two experiments?
- Is the collected sample a representation of the population?
- Is the result obtained significant enough to make a difference?



Define who matters, gather observations, describe the sample, then reason about the population.

A decision follows the evidence, with uncertainty in view.

Population

The population is the entire group we want to study or draw conclusions about.

- All students at a university
- All customers of Amazon
- All registered voters in North Carolina
- All company transactions in 2026



The group can be people, transactions, devices, events, or other units.

Population size is often written as N.

Sample

A sample is the subset of a population that we actually observe.

Research question

How many hours do UNCG students sleep each night?

Population

All UNCG students



500 randomly selected students

We observe the sample to learn about the population.

Why Do We Use Samples?

- 01 Cost** A census may be too expensive.
- 02 Time** Collecting every observation can take too long.
- 03 Scale** The population may be extremely large.
- 04 Change** The population may change continuously.
- 05 Access** Measuring everyone may be impossible.

Example: Netflix viewing behavior

Hundreds of millions of users



Analyze a representative sample



Question to keep in mind

Does the sample represent the population?

Quick Exercise: Population or Sample?

A university wants the average weekly study time of its 20,000 students.

Researchers randomly survey 600 students.

- 1 What is the population?
- 2 What is the sample?
- 3 What issue arises if researchers survey only students in the library?

Check

Population: all 20,000 students | Sample: 600 surveyed students | Library-only survey: possible sampling bias

What Is a Variable?

A variable is a characteristic that can take different values across observations.

Student	Study hours	Attendance	Major	Exam score
Alice	8	90%	CS	86
Bob	4	75%	Biology	72
Carol	10	95%	CS	93

Variables

Study hours

Attendance

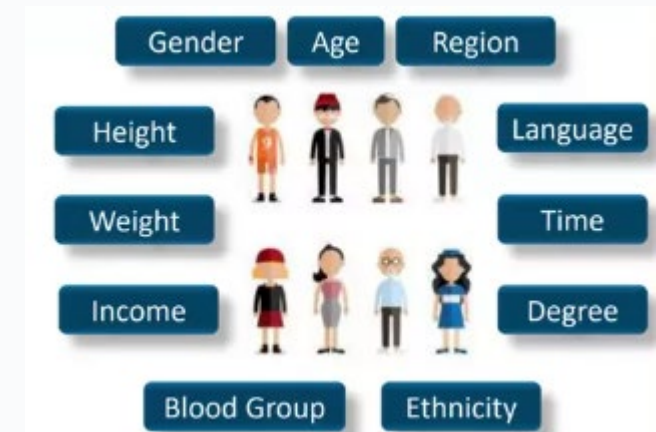
Major

Exam score

One row = one observation

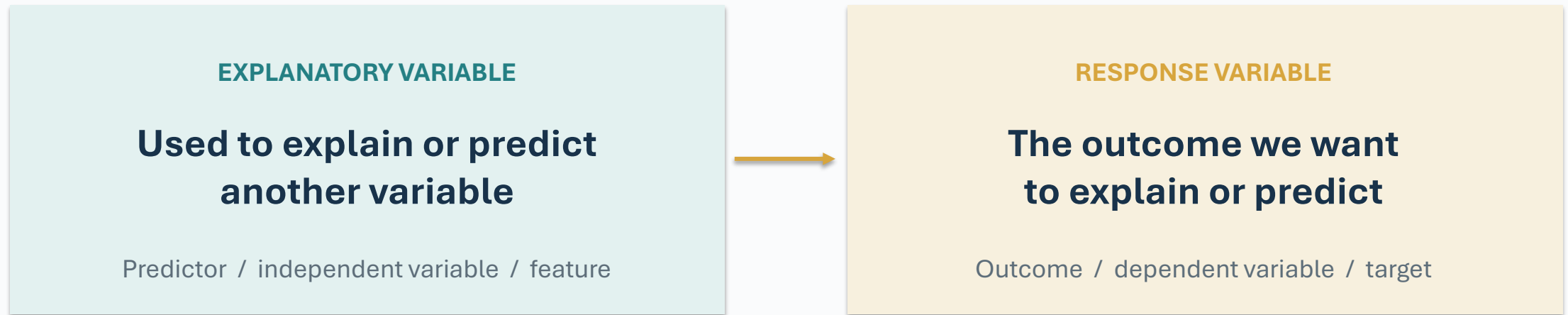
One column = one variable

In a pandas DataFrame: rows are observations; columns are variables.



Explanatory and Response Variables

When we study a relationship, variables can play different roles.

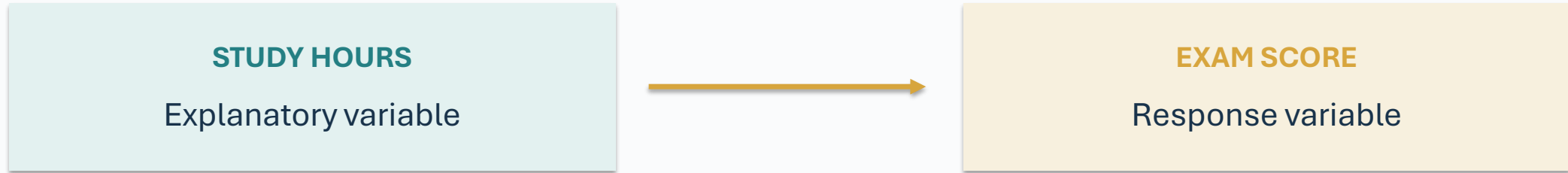


These labels describe the roles variables play in a particular question.

number_project	average_monthly_hours	time_spend_company	Work_accident	left	promotion_last_5years	dept	salary
2	157	3	0	1	0	sales	low
5	262	6	0	1	0	sales	medium
7	272	4	0	1	0	sales	medium
5	223	5	0	1	0	sales	low
2	159	3	0	1	0	sales	low

Example: Study Time and Exam Performance

Research question: Does study time help predict exam performance?



Try another

Can house size predict house price?

Which variable explains? Which outcome do we predict?

Association Does Not Prove Causation

Students who study more often earn higher grades.

An association appears in the data.

Does more study time cause higher grades?

Prior knowledge

Motivation

Attendance

Course difficulty

Other factors may influence study time and grades.

Keep this distinction in mind when interpreting correlation or regression.

Two Major Types of Variables

QUANTITATIVE

Numbers represent amounts or measurements.

Age
Height
Income
Temperature
Exam score

QUALITATIVE

Values represent categories or groups.

Major
Eye color
State
Payment method
Product category

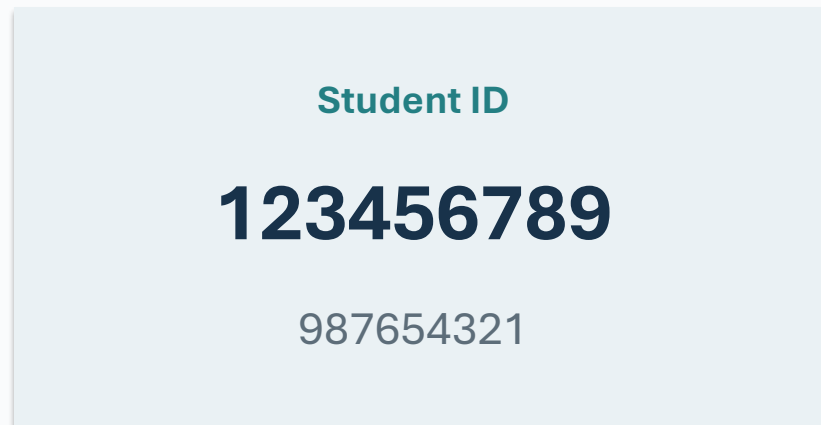
number_project	average_monthly_hours	time_spend_company	Work_accident	left	promotion_last_5years	dept	salary
2	157	3	0	1	0	sales	low
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Quantitative

Qualitative

Numbers Can Be Labels

A numeric format does not guarantee quantitative data.



Average ID?

$$(123456789 + 987654321) / 2$$



Meaningful?

No. IDs are labels, not amounts.

Other numeric labels

ZIP code

Phone number

Jersey number

Exercise: Identify the Variable Type

For each variable, choose quantitative or qualitative.

01 Age

02 Major

03 GPA

04 ZIP code

05 Number of siblings

06 Satisfaction: Poor / Fair / Good / Excellent

07 Student ID

Satisfaction categories have a natural order. The distinction has a name: ordinal data.

Exercise: Identify the Variable Type

number_project	average_monthly_hours	time_spend_company	Work_accident	left	promotion_last_5years	dept	salary
2	157	3	0	1	0	sales	low
5	262	6	0	1	0	sales	medium
7	272	4	0	1	0	sales	medium
5	223	5	0	1	0	sales	low
2	159	3	0	1	0	sales	low

	Sqft	Floor	TotalFloor	Bedroom	Living.Room	Bathroom	Price
1	1177.698	2	7	2	2	2	62000
2	2134.800	5	7	4	2	2	78000
3	1138.560	5	7	2	2	1	58000
4	1458.780	2	7	3	2	2	45000
5	967.776	11	14	3	2	2	45000

Binary Variables

Categorical Variables

Ordinal Variables

Continues Variables

Discrete Variables

Parameter vs. Statistic

Both are numerical summaries. The difference is which group they describe.

POPULATION

Parameter

Describes the entire population

μ σ^2 σ

population mean, variance, standard deviation



SAMPLE

Statistic

Calculated from a sample

$\bar{x}$ s^2 s

sample mean, variance, standard deviation

Population → parameter | Sample → statistic

Example: Parameter vs. Statistic

UNCG has 20,000 students.

True average height

$$\mu = 169.2 \text{ cm}$$

Parameter: describes all students

Random sample of 100

$$\bar{x} = 168.7 \text{ cm}$$

Statistic: calculated from the sample

Use the statistic to estimate the parameter.

The sample value will usually differ somewhat from the population value.

Two Major Branches of Statistics

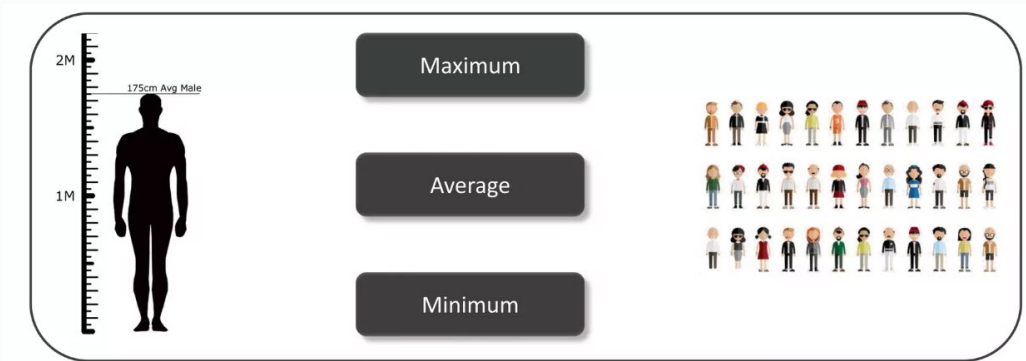
The same dataset can support description and inference.

DESCRIPTIVE STATISTICS

Uses the data to provide descriptions of the population, either through numerical calculations or graphs or tables.

Mean
Median
Standard deviation
Tables and histograms

Question: What does this dataset look like?

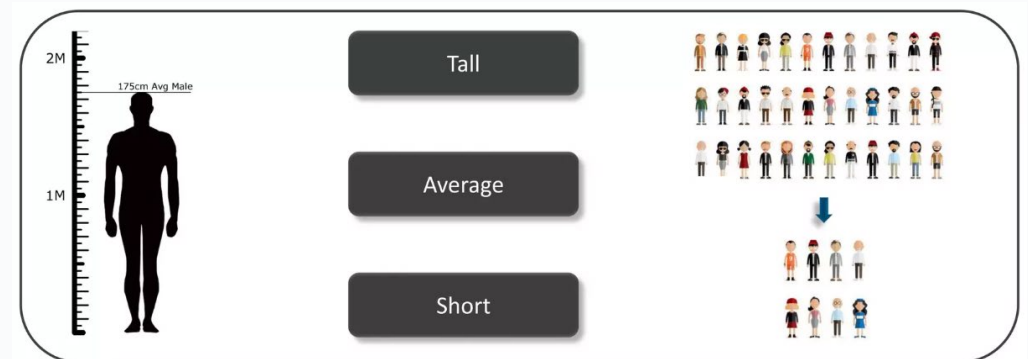


INFERENCE STATISTICS

makes inferences and predictions about a population based on a sample of data taken from the population in question.

Estimation
Confidence intervals
Hypothesis tests
Regression

Question: What can this sample tell us?



A Sample Result and a Population Claim

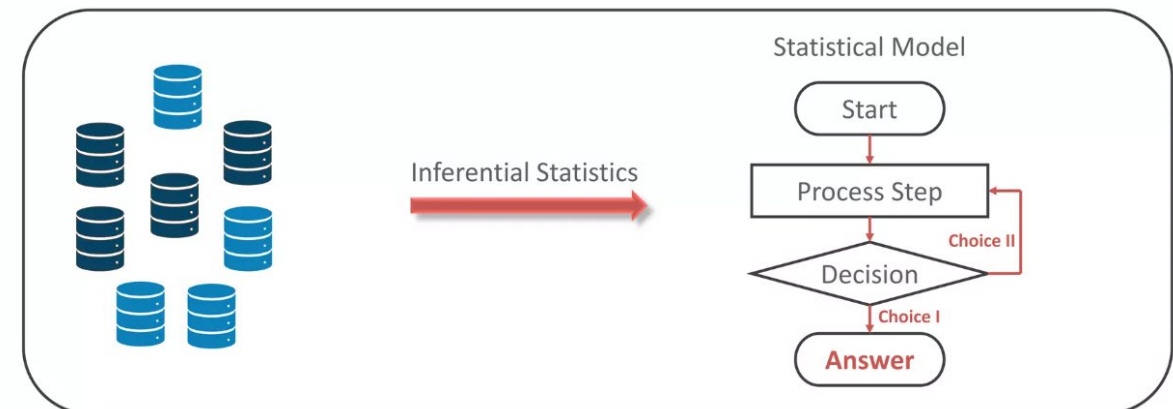
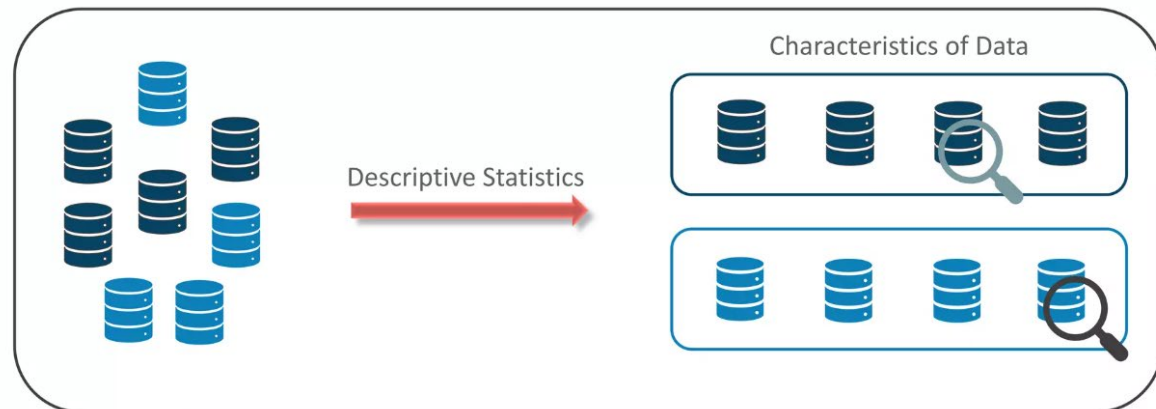
A survey of 500 UNCG students finds an average of 6.7 hours of sleep per night.

DESCRIPTIVE

“The 500 surveyed students sleep an average of 6.7 hours.”

INFERENTIAL

“UNCG students sleep about 6.7 hours per night.”



How Can We Describe a Typical Value?

Three common measures of center answer the question in different ways.

MEAN

Arithmetic average

$$\bar{x} = \sum x_i / n$$

MEDIAN

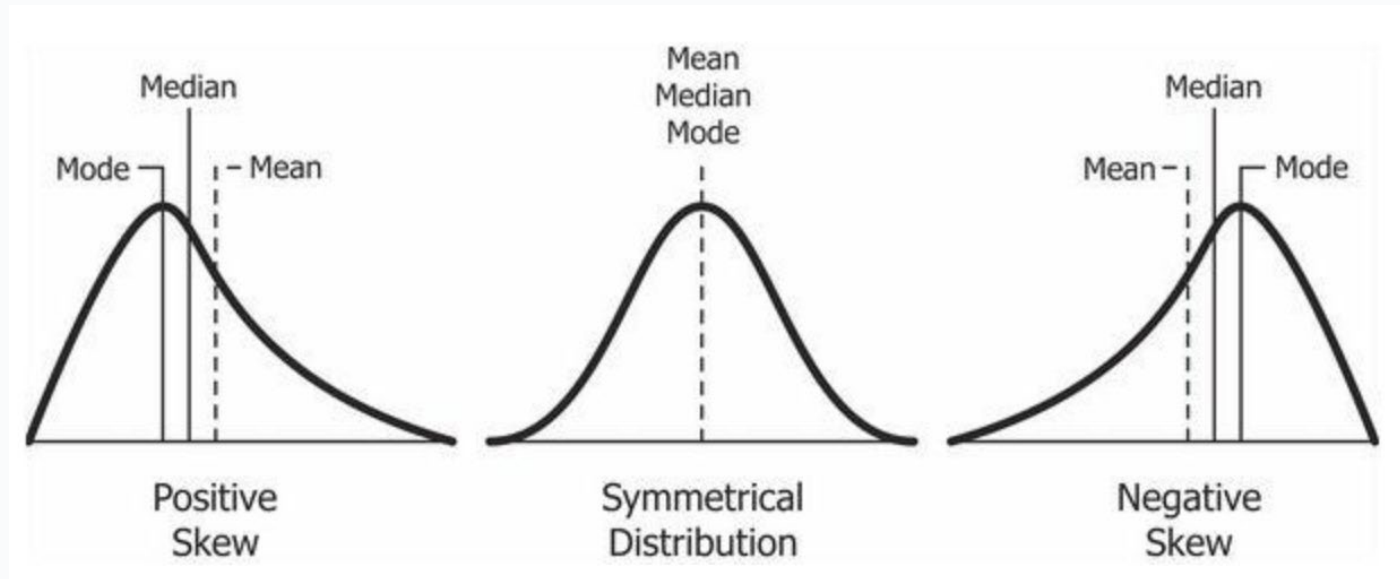
Middle value after sorting

Middle observation

MODE

Most frequent value

Value that appears most



Mean, Median, and Mode: Example

Dataset

2, 3, 3, 4, 5, 5, 5, 6, 30

7

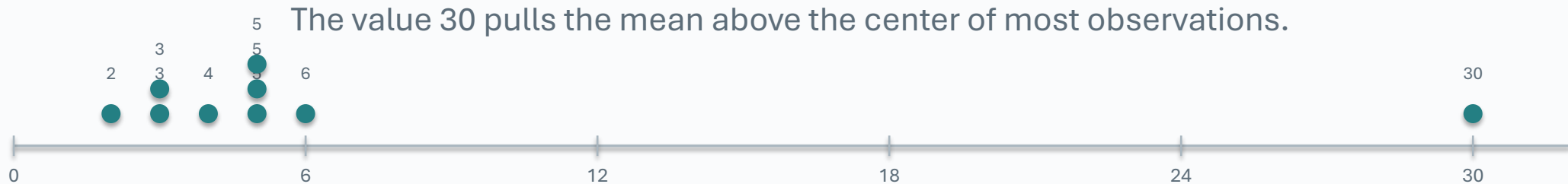
Mean = $63 / 9$

5

Median

5

Mode

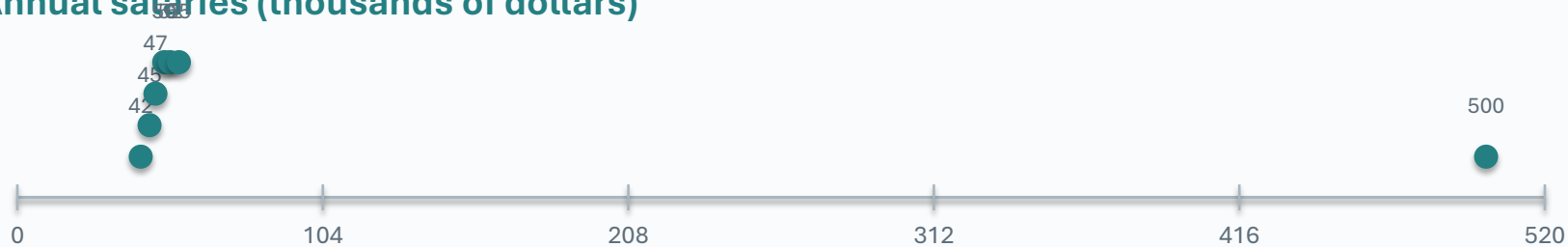


Mean or Median?

Mean often works well for symmetric data without extreme outliers.

Median often works better for skewed data or when extreme values exist.

Annual salaries (thousands of dollars)



42 45 47 50 52 55 500

The \$500K salary pulls the mean far above the middle value.

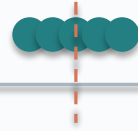
Which measure would you report as a typical salary?

The Center Alone Does Not Describe the Data

Both classes have a mean of 80.

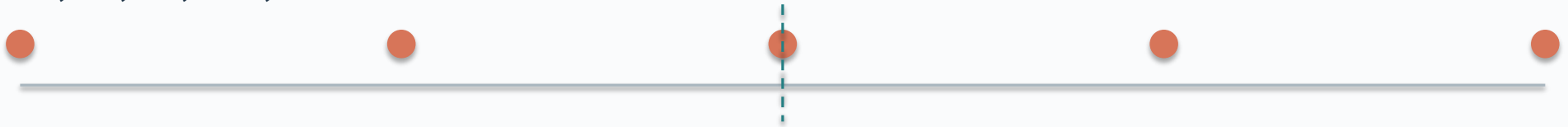
Class A

78, 79, 80, 81, 82



Class B

40, 60, 80, 100, 120



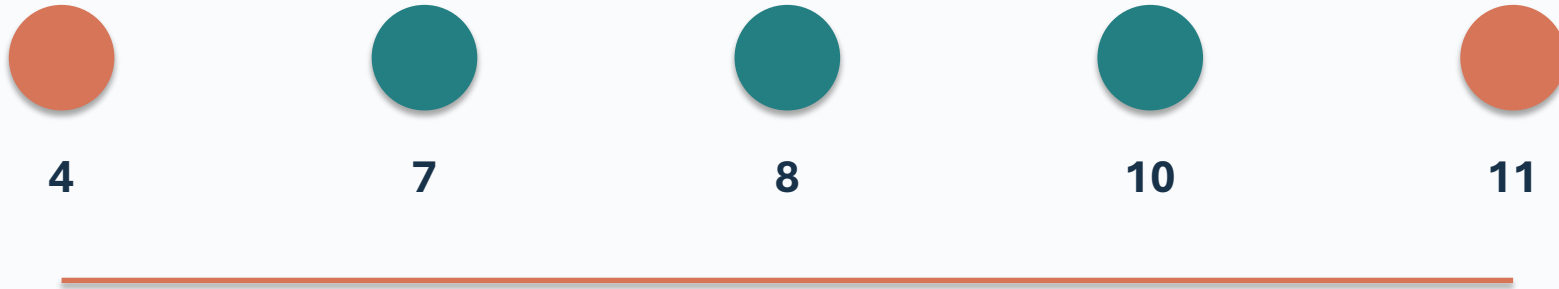
Same center. Very different spread.

We also need measures of variability.

Range

Range = maximum – minimum

Example: 4, 7, 8, 10, 11



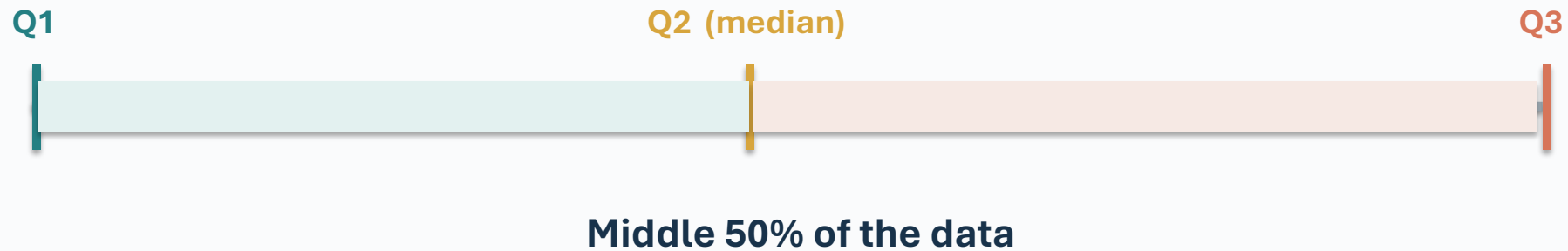
$$11 - 4 = 7$$

Easy to calculate

Uses only two observations and is sensitive to outliers

Interquartile Range (IQR)

Quartiles divide ordered data into four parts.



$$\text{IQR} = Q3 - Q1$$

Less sensitive to extreme values than the range.

Outlier fences: $x < Q1 - 1.5(\text{IQR})$ or $x > Q3 + 1.5(\text{IQR})$

Why Square the Deviations?

Start with the data 2, 4, 6. The mean is $\bar{x} = 4$.

x	$x - \bar{x}$	$(x - \bar{x})^2$
2	-2	4
4	0	0
6	+2	4

Problem

$$-2 + 0 + 2 = 0$$

Positive and negative deviations cancel.

Square each deviation

$$4 + 0 + 4 = 8$$

Squaring prevents cancellation and weights larger gaps more.

Population Variance

Variance measures the average squared distance from the population mean.

$$\sigma^2 = \sum (x_i - \mu)^2 / N$$

x_i each observation

μ population mean

N population size

Large variance

Values are more spread out.

Small variance

Values are more concentrated around μ .

Sample Variance

For a sample, calculate squared deviations from the sample mean.

$$s^2 = \sum (x_i - \bar{x})^2 / (n - 1)$$

Why divide by $n - 1$?

The sample mean comes from the same sample. Dividing by $n - 1$ corrects a tendency to underestimate population variability.

Bessel's correction

We will not derive it today.



Bessel's correction

$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n} \times \frac{n}{n-1} = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$

Annotations:

- x_i : i^{th} data point
- $\bar{x}$: sample mean
- s^2 : sample variance
- n : sample size
- $\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$: Biased estimate of population variance
- $\frac{n}{n-1}$: Bessel's correction factor
- $\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$: Unbiased estimate of population variance

Standard Deviation

Variance is expressed in squared units. If height is measured in cm, variance is in cm^2 .

Take the square root to return to the original unit.

Population standard deviation

$$\sigma = \sqrt{\sigma^2}$$

Sample standard deviation

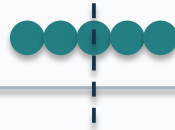
$$s = \sqrt{s^2}$$

Interpreting Standard Deviation

Both datasets have a mean of 50.

Dataset A

48, 49, 50, 51, 52



Dataset B

20, 35, 50, 65, 80



Dataset B has greater variability, so $SD_B > SD_A$.

The points sit farther from the mean.

Integrated Challenge: A Small Student Dataset

Student	Major	Study hrs	Sleep hrs	Exam score
A	CS	3	8	68
B	CS	5	7	75
C	Biology	6	7	81
D	Business	7	6	85
E	CS	8	5	89
F	Biology	12	4	94

Work through these questions

- 1 Which variables are qualitative? Which are quantitative?
- 2 If study time predicts exam performance, what are the explanatory and response variables?
- 3 Does this sample prove that studying more causes higher scores?

Challenge: Salaries with an Outlier

Annual salaries (thousands of dollars)

42, 45, 47, 48, 50, 52, 55, 60, 350

- 1 Calculate the mean and median.
- 2 Which better represents a typical salary?
- 3 Would you use range, IQR, or standard deviation to summarize spread?

Think about how the extreme salary affects each measure.

Challenge: Same Mean, Different Reliability

Two servers report response times in milliseconds.

Server A 98, 99, 100, 101, 102

Server B 60, 80, 100, 120, 140

-
- 1 Calculate each mean and range.
 - 2 Which server would you choose for a web service, and why?
 - 3 Which measure captures that difference?

Challenge: From Sample to Population

A streaming company has 10 million users.

It randomly selects 20,000 users and reports, “Our users watch 2.4 hours per day on average.”

10 million

Population

20,000

Sample

$\bar{x} = 2.4$ hr/day

$s = 1.1$ hours

- 1 What is the population and sample?
- 2 Is 2.4 a parameter or statistic?
- 3 Is the company’s statement descriptive or inferential?
- 4 Can the population mean be exactly 2.4 hours?